

Absolute Market Unfairness and the Compression of Bid–Ask Spreads in Cryptocurrency Option Markets

Balázs Králik*, Nóra Felföldi-Szűcs†, Tamás Rutai‡, Kata Váradi§

Abstract

Put–call parity ties a European call, a put, and a matching future into a one-to-one correspondence, so on a single venue their standing quotes should be mutually consistent up to transaction costs. We propose *Absolute Market Unfairness* (AMU): the part of the quoted put–call-parity violation that remains, at the top of the book, after netting out direct transaction costs. Unlike the put–call-parity literature, which tests *trade* prices, AMU is a statement about *quotes*. It is a model-free, incontrovertible symptom of poor liquidity provision—comparable in basis points across venues, underlyings, and maturities—in the spirit of the best-execution (NBBO) rules that police quote quality where identical instruments trade frictionlessly on several venues. It does not fully characterize a market’s liquidity, but it gives a clean benchmark for comparison. Using high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching futures over 2020–2026, we describe the level and cross-section of AMU and its evolution through the 2024 regulatory changes (the U.S. spot Bitcoin and Ether ETFs and MiCA). AMU differs only moderately across venues and underlyings; over the option surface it varies mainly through how *often* a wedge appears rather than how large it is; and it compressed markedly around 2024, robustly to the assumed cost level. Panel regressions with cell fixed effects show the compression persists after controlling for spreads, depth, and quote staleness. We read this as descriptive evidence consistent with the entry of participants whose quoting improved market quality, rather than a mere increase in passive depth; the design is not causal.

Keywords: cryptocurrency options, bid–ask spreads, absolute market unfairness, put–call parity, limits to arbitrage, ETF approval, MiCA

JEL classification: G12, G14, G15, G23

Contents

1 Introduction

2

*Corresponding Author. Assistant Professor at Corvinus University of Budapest, Institute of Finance, Fővám square 8. Budapest, Hungary, 1093. E-mail: balazs.kralik@uni-corvinus.hu.

†Associate Professor at Corvinus University of Budapest, Institute of Finance.

‡Student at Corvinus University of Budapest.

§Associate Professor at Corvinus University of Budapest, Institute of Finance.

2	Related literature	3
2.1	Put–call parity and crypto derivative efficiency	3
2.2	Market quality, bid–ask spreads, and the limits of arbitrage	4
3	The absolute market unfairness: concept and measurement	4
3.1	Market maker advantage and the absolute market unfairness	5
3.2	The range of arbitrage configurations	6
3.3	The no-AMU band and the shape of the MMA distribution	7
3.4	Data and sample	9
3.5	Panel construction	9
4	The level and cross-section of the absolute market unfairness	11
4.1	The shape of the MMA distribution and why it peaks below zero	11
4.2	RQ1: AMU across exchanges, moneyness, and maturity	11
5	The 2024 regulatory shift	12
6	Did the absolute market unfairness fall in 2024?	15
6.1	Time-series and distributional evidence	15
7	Regression analysis	17
7.1	Empirical design	17
7.2	Results	20
8	Conclusion	22

1 Introduction

This paper studies the top-of-book quotes of cryptocurrency option markets, and the premium that market makers require to quote them. The object is not trade prices, but standing quotes: the top of the book is a public good, and its quality is a statement about the market’s efficiency. The bid-ask spread is one of the most important frictions that impede the law of one price. TODO: cite something.

There is no general theory of how wide a spread should be, but it is understood that it is compensation for the market maker to provide liquidity (TODO: cite something). How much compensation is fair is a matter of what model we choose for the behavior of market makers (TODO: cite something). However, there are some special cases where we CAN identify clearly unfair quotes. The best known is the case where identical instruments are quoted on several venues with no friction between transacting on one or another: this the the Nasdaq. In this case the NBBO rule explicitly forbids filling retail market order flow at a price worse than the across-ecn best price.

There are other situations which might seem similar: The ETF and ADR markets, as well as the On/Off-the-run Treasury markets (TODO cite papers relating to these). But here there are real frictions (liquidity and market access) that legitimately keep the ETF market price vs ETF creation/redemption cost, or ADR and equity price, or on/off-the-run Treasury price, from being equal.

In this paper we show a situation where there are equivalent instruments traded on a single market, where price differentials should not exist at all: These are option markets in european options with matching futures. The put-call parity (PCP) relationship establishes

a one-to-one correspondence between a put and a future and a call. The breaking of the PCP has been extensively studied in various markets (TODO: put in citations), and it has been shown that most apparent PCP breaking is due to transaction costs involved in replicating a call from a put. PCP breaking remaining beyond what can be explained by transactions costs are assumed to be idiosyncratic.

We propose a to use the PCP breaking remaining in quotes after accounting for transaction costs as a measure of the quality of a market or venue for particular option chains. This measure we propose to call AMU (Absolute Market Unfairness), because it can be easily interpreted to represent an incontrovertible symptom of a market with poor liquidity provision. AMU is not meant to characterize a venue’s liquidity completely, but it does provide a model-free measure for comparison.

Our paper will focus on characterizing AMU on cryptocurrency option markets. We describe empirical facts about AMU, including its response to regulatory changes in 2024. We find evidence corroborating the fact that the regulatory changes have introduced new players whose market making behavior improves quoting quality, as opposed to just introducing more liquidity.

The empirical analysis is organized around three research questions:

1. How large is AMU, and does it differ across exchanges and underlyings?
2. Did AMU drop significantly upon the 2024 regulatory and institutional regime shift?
3. Is any post-2024 drop fully explained by the usual liquidity frictions, or does it partly reflect a lower premium demanded by a new class of parity-enforcing participants?

We answer all three. Section 2 places AMU in the literature. Section 3 states the cost-plus-premium identity, defines MMA and AMU, and describes the data and panel. Section 4 presents the level and cross-section of AMU. Section 5 describes the 2024 regulatory shift. Section 6 documents the compression and its robustness to the cost assumption. Section 7 reports the regressions. Section 8 concludes.

2 Related literature

2.1 Put–call parity and crypto derivative efficiency

Parity has long been tested, but on trade prices, not quotes.

Testing put–call parity has a long tradition in equity, index, and currency options. Early work shows apparent violations are mostly explained by bid–ask spreads, execution timing, short-sale constraints, taxes, and trading costs (Klemkosky & Resnick 1979, Nisbet 1992, Kamara & Miller 1995, El-Mekkaoui & Flood 1998, Cremers & Weinbaum 2010). Crypto options add complications: contracts are often inverse or quanto, collateral and settlement currencies differ from payoff currencies, and the reference index is not always tradable (Alexander et al. 2023, Lucic & Sepp 2024). This literature tests trade or closing prices. Our object is the standing quotes.

Recent crypto work studies efficiency; we add a quote-level measure.

Recent work treats derivative efficiency as a market-structure problem. Alexander et al. (2024) link Bitcoin and Ether options to perpetual futures and find efficiency improving over time, especially at long maturities. Felföldi-Szűcs (2024) study high-frequency parity in Binance Ether options and show violations shrink once synchronization and costs are

handled. We extend this to inverse options and matching futures on Deribit and OKX, and add a comparable basis-point measure of the quoted parity gap.

2.2 Market quality, bid–ask spreads, and the limits of arbitrage

Arbitrage is costly, so a no-arbitrage band opens.

Reading AMU as a symptom of costly, limited arbitrage rests on a simple idea: arbitrage is costly and capital constrained, so a band opens and only wedges that pierce it are worth trading. Amihud & Mendelson (1986) show spreads affect prices because investors require compensation for illiquidity. Shleifer & Vishny (1997) stress that textbook arbitrage is neither riskless nor free in practice. Gârleanu & Pedersen (2011) show how margin and funding constraints create deviations from the law of one price. All of this applies to crypto, where collateral, margin, exchange credit risk, and stablecoin conversion raise the cost of enforcing parity.

Required mispricing splits into direct cost plus a residual.

The split of required mispricing into direct costs plus a residual is standard. Pontiff (1996, 2006) shows holding costs—including the idiosyncratic risk of an undiversified arbitrageur—dominate, and mispricing survives in proportion to them. Chen et al. (2021) estimate the no-arbitrage band directly from data. Gromb & Vayanos (2010) survey the theory: frictions that make convergence risky widen the band. For the market maker, Bollen et al. (2004) split the quoted spread into order processing, inventory holding, and adverse selection. AMU is the analogous residual for the parity quote—the part we can call clearly unfair without a model of fair compensation.

Such bands persist and move with market structure elsewhere.

No-arbitrage bands persist and move with market structure in other markets too. Balance-sheet constraints produced persistent covered interest parity deviations (Du et al. 2018); ETF premia persist when the underlying is illiquid or arbitrageurs are constrained (Pan & Zeng 2017, Todorov 2021). Crypto runs the other way: regulatory clarity and institutional access plausibly *lowered* the cost of arbitrage capital over our sample, which would compress the band.

3 The absolute market unfairness: concept and measurement

The organizing idea is the accounting identity of costly arbitrage:

$$\text{required mispricing} = \text{direct transaction costs} + \text{absolute market unfairness}. \quad (1)$$

AMU is the parity gap left in quotes after direct costs.

A quoted parity deviation is worth acting on only once it covers the direct costs of the round trip. Anything left over is the residual we call AMU. The displaced object is a *quote*: a top-of-book price sitting away from its parity-implied benchmark. We do not decompose that residual. It may reflect a risk premium (counterparty, settlement, convergence, inventory, funding, adverse selection), latency, stale books, tick-size effects, or a low chance of a fill; the limits-to-arbitrage literature catalogues such frictions (Pontiff 2006, Chen et al. 2021, Gromb & Vayanos 2010), and the spread decomposition names their market-making counterparts (Bollen et al. 2004). Our claim is narrower and model-free: whatever its source, a positive AMU is a cost-adjusted parity gap left in the quotes, hence a

symptom of imperfect quoting. Everything downstream is net of one direct-cost parameter (0.003 of notional); Section 6 shows the conclusions do not depend on it.

3.1 Market maker advantage and the absolute market unfairness

Parity in crypto options must be written after currency conversion.

The benchmark is still put–call parity, but implementation is contract-specific. Our most liquid contracts—inverse BTCUSD and ETHUSD options on Deribit and OKX—are quoted against a USD index, while collateral, premium, and settlement are in coin. Parity must therefore be written after converting premia and the futures leg to a common currency (Alexander et al. 2023, Lucic & Sepp 2024): the coin-denominated premia are scaled by the conversion price S , and contract size cancels in basis points of notional.

Four paths implement the arbitrage.

For a matched strike–expiry pair the profit depends on how the arbitrage is run: forward or backward, joining the call or the put side. In every path the maker *joins* a standing quote rather than crossing it: he posts and profits only if a taker hits him. For instance, in forward/join-call he buys the call at C_{bid} ; in backward/join-put he sells it at C_{bid} . This gives four paths (Table 1), each net of a round-trip cost and converted to basis points by $10,000/\text{index_level}$.

A tickpath is one (pair, path) observation.

Let $m_{i,p,t}$ be the signed basis-point payoff of path p for pair i at time t , after conversion and cost adjustment. We call each (pair, path) observation a *tickpath*: every synchronized quote (“tick”) for a matched pair gives four tickpaths, one per path, so a cell (g, t) with $n_{g,t}$ pairs has $4n_{g,t}$ tickpaths. Both objects below are averages of m over tickpaths; they differ only in which tickpaths enter.

Market maker advantage (MMA) is the mean of $m_{i,p,t}$ taken over *all* tickpaths in a cell (g, t) :

$$\text{MMA}_{g,t} = \frac{1}{4n_{g,t}} \sum_{i \in g,t} \sum_{p=1}^4 m_{i,p,t}, \quad (2)$$

where $n_{g,t}$ is the number of matched pairs in the cell. The average includes the many paths with no positive profit after costs, so *mean MMA is negative*: it sits near minus the round-trip cost. MMA is the outcome of a maker who joins every quote indiscriminately.

The absolute market unfairness (AMU) comes in two complementary forms, both built from the positive part of m . The *conditional* AMU averages m over only the *positive-MMA tickpaths*—those with $m_{i,p,t} > 0$:

$$\text{AMU}_{g,t}^c = \frac{\sum_{i \in g,t} \sum_{p=1}^4 \max(0, m_{i,p,t})}{\sum_{i \in g,t} \sum_{p=1}^4 \mathbf{1}\{m_{i,p,t} > 0\}}, \quad (3)$$

the average *size* of the wedge, given that one exists. Let the *positive-wedge frequency* $\phi_{g,t}$ be the share of the $4n_{g,t}$ tickpaths with a positive payoff,

$$\phi_{g,t} = \frac{1}{4n_{g,t}} \sum_{i \in g,t} \sum_{p=1}^4 \mathbf{1}\{m_{i,p,t} > 0\}, \quad (4)$$

and let the *unconditional* AMU average the same positive part over *all* tickpaths:

$$\text{AMU}_{g,t}^u = \frac{1}{4 n_{g,t}} \sum_{i \in g,t} \sum_{p=1}^4 \max(0, m_{i,p,t}) = \phi_{g,t} \text{AMU}_{g,t}^c. \quad (5)$$

The two forms factor into frequency times size.

The identity $\text{AMU}^u = \phi \text{AMU}^c$ is the paper’s organizing decomposition: the per-quote wedge is how *often* a wedge appears (ϕ) times how *large* it is when it does (conditional AMU). We read the cross-section through this size/frequency split (Figures 5–6); the time series and regressions use the unconditional AMU. Both are in basis points of notional.

AMU is the expected profit of an imaginary maker who joins only the favorable side.

Read AMU through an imaginary maker who joins a quote only when the parity payoff is in his favor. It is an expectation, not a realized profit: whether he is filled depends on queue position and on a taker crossing. The bid–ask spread already pays him for the usual costs of making a market—order processing, inventory, adverse selection, queue risk. AMU is separate: it measures only how differently the two equivalent instruments—the real quote and its parity-implied synthetic—are priced against one another. MMA, conditional AMU, and unconditional AMU are three summaries of the same tickpath distribution of m : its mean over all tickpaths, its mean over the positive ones, and the mean of its positive part.

Table 1: Arbitrage-profit formulas by path and contract style. To obtain the basis-point statistics used for MMA/AMU, multiply each expression by 10,000/index_level. Each expression includes the trading-cost term $\text{cost} = \text{cost_per_notional} (0.003) \times \text{index_level}$ from Table 3. Contract size cancels in the basis-point expression. For inverse contracts, S_{bid} and S_{ask} are the prices at which the coin payoff is converted to USD or stablecoin.

Path	Linear contracts	Inverse contracts
fwd, join call	$C_{bid} - P_{bid} - (F_{bid} - K)e^{-rT} - \text{cost}$	$(C_{bid} - P_{bid})S_{bid} - (F_{bid} - K)e^{-rT} - \text{cost}$
fwd, join put	$C_{ask} - P_{ask} - (F_{bid} - K)e^{-rT} - \text{cost}$	$(C_{ask} - P_{ask})S_{ask} - (F_{bid} - K)e^{-rT} - \text{cost}$
bck, join call	$P_{ask} - C_{ask} + (F_{ask} - K)e^{-rT} - \text{cost}$	$(P_{ask} - C_{ask})S_{ask} + (F_{ask} - K)e^{-rT} - \text{cost}$
bck, join put	$P_{bid} - C_{bid} + (F_{ask} - K)e^{-rT} - \text{cost}$	$(P_{bid} - C_{bid})S_{bid} + (F_{ask} - K)e^{-rT} - \text{cost}$

The cost assumption only sets the scale.

Trading costs are knowable but venue-dependent and messy. We assume a fixed proportional cost of 0.003 per round trip. Two things make this harmless. First, the pre/post dynamics of unconditional AMU barely move with the cost level, which we check over a cost grid (Figure 11); the conditional form is more cost-sensitive, one reason the headline results use the unconditional measure. Second, a fixed cost across venues keeps cross-exchange comparisons coherent. The cost only sets AMU’s scale.

3.2 The range of arbitrage configurations

Geometrically, AMU compares a real quote to its synthetic twin.

It is useful to see what AMU is measuring geometrically. For a given option leg—say the call—the quotes define a *real* market $[C_{bid}, C_{ask}]$, while parity combined with the put and the matching futures defines a *synthetic* market $[C_{bid}^{\text{syn}}, C_{ask}^{\text{syn}}]$, where $C^{\text{syn}} = P + (F - K)e^{-rT}$.

Both intervals live on the same price axis, and their relative position determines whether, and how strongly, an arbitrage exists. Figure 1 shows the three qualitatively distinct configurations.

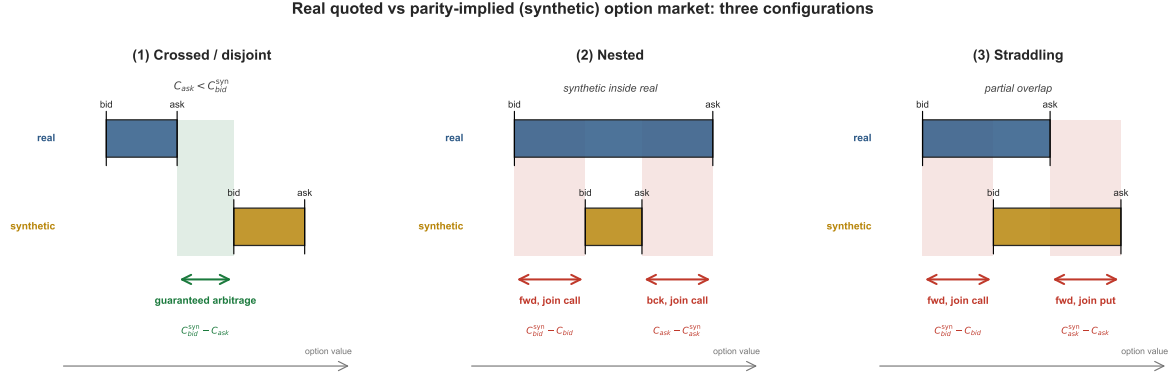


Figure 1: Real quoted versus parity-implied (synthetic) option market. When the markets are crossed and disjoint ($C_{ask} < C_{bid}^{syn}$, or the symmetric case), there is a single guaranteed riskless arbitrage. When one market is nested inside the other, or when the two straddle each other, there is no guaranteed profit but two paths carry a *potential* market-maker advantage, realized only if the maker’s quote is joined and then hit. In the nested case these are the forward and backward join-call paths ($C_{bid}^{syn} - C_{bid}$ and $C_{ask} - C_{ask}^{syn}$), with the join-put payoffs equal to their negatives (exactly so when the futures market has zero spread; the trading cost need not be zero); in the straddling case they are the forward join-call and forward join-put paths ($C_{bid}^{syn} - C_{bid}$ and $C_{ask}^{syn} - C_{ask}$). The latter two configurations are what AMU aggregates. Schematic, illustrative levels.

Only the crossed case is a guaranteed arbitrage.

In the crossed case the real ask lies below the synthetic bid (or vice versa); buying one and selling the other locks in the gap. In the nested and straddling cases there is no guaranteed profit: the payoff depends on which side is joined and whether a taker crosses. These conditional states are what MMA and AMU aggregate over the surface.

AMU is not the spread; it is where the quote sits.

MMA and AMU do not measure the quoted half-spread itself—the compensation makers demand for liquidity provision. They measure where the quotes *sit*: each real quote must track its synthetic counterpart within the cost cushion (Section 3.3), and AMU is what stands outside that tolerance. This is the through-price gap of the NBBO analogy from the introduction, with the parity-implied synthetic market as the second venue.

3.3 The no-AMU band and the shape of the MMA distribution

How far can a quote move before it is unfair?

The configurations in Figure 1 raise a sharper question: for a *given* synthetic market, how far can a real quote move before it becomes unfair? The answer pins each real quote to its synthetic counterpart within a tolerance set by the trading cost, and it fixes the shape that the MMA distribution “should” take.

Two paths bound the fair call ask.

Consider the real call ask C_{ask} . Two of the four paths in Table 1 contain it—forward/join-

put and backward/join-call. Requiring both payoffs to be non-positive (no AMU) yields

$$P_{ask} + (F_{ask} - K)e^{-rT} - \text{cost} \leq C_{ask} \leq P_{ask} + (F_{bid} - K)e^{-rT} + \text{cost}. \quad (6)$$

A fair quote must track its synthetic within a cost cushion.

Both bounds use the *same-side* put quote P_{ask} , so the band is centred on the synthetic ask at the futures mid, $C_{ask}^{\text{syn}} \equiv P_{ask} + (F_{mid} - K)e^{-rT}$, with half-width $\text{cost}' \equiv \text{cost} - \frac{1}{2}(F_{ask} - F_{bid})e^{-rT}$. The symmetric calculation pins C_{bid} to C_{bid}^{syn} . So a fair real market must track the synthetic market quote by quote, to within $\pm \text{cost}'$ (Figure 2). The minus sign in cost' is correct: when the futures market is very wide, cost' turns negative, meaning no fair synthetic market exists and AMU is undefined. This did not occur in our data.

The band's width comes from the cost.

The role of the cost is easy to mis-state. It is *not* that C_{ask} merely lies inside the synthetic bid-ask; that would carry no cost at all. Rather, C_{ask} is tied to the synthetic *ask* within the $\pm \text{cost}'$ cushion. With $\text{cost} = 0$ and a positive futures spread, cost' is negative, so the interval in (6) inverts and is empty: no strictly fair ask exists. As all frictions vanish, the band collapses and the real market should become strictly identical to the synthetic one—the law of one price.

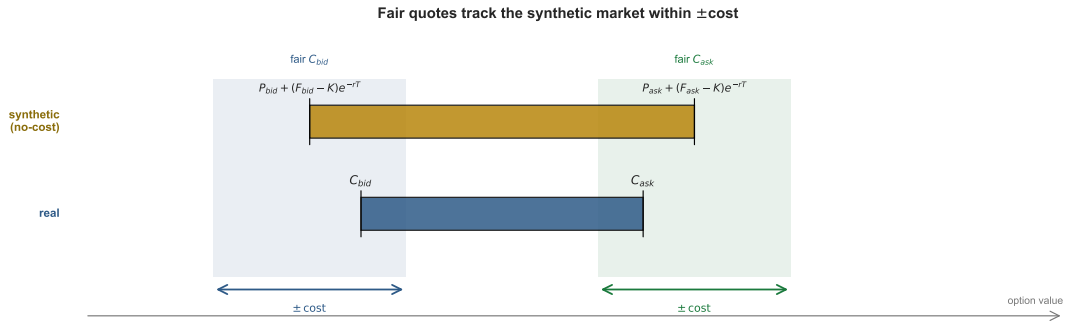


Figure 2: No AMU requires each real quote to track its matching synthetic quote within $\pm \text{cost}'$ (the half-width $\text{cost}' \equiv \text{cost} - \frac{1}{2}(F_{ask} - F_{bid})e^{-rT}$). The band centres are the frictionless (no-cost) synthetic prices $P_{bid} + (F_{bid} - K)e^{-rT}$ and $P_{ask} + (F_{ask} - K)e^{-rT}$; the cushion around each is $\pm \text{cost}'$. As frictions shrink, the real market is squeezed onto the synthetic market. Schematic, illustrative levels.

Uniform quoting would imply a flat MMA rectangle.

Equation (6) also fixes the shape of the MMA distribution. Write $u = C_{ask} - C_{ask}^{\text{syn}}$; the two conjugate C_{ask} payoffs are exactly $u - \text{cost}'$ (forward, join put) and $-u - \text{cost}'$ (backward, join call). As C_{ask} ranges over its fair band $u \in (-\text{cost}', \text{cost}')$, each sweeps linearly over $(-2\text{cost}', 0)$. If makers quoted with no preferred point inside the band, the implied MMA would be *uniform* on $(-2\text{cost}', 0)$: a rectangle of width $2\text{cost}'$ just left of zero. That rectangle is the no-information benchmark (dashed in Figure 3).

Competition instead concentrates MMA near minus the cost.

The observed distribution instead peaks sharply near $-\text{cost}'$. Two things point there, and it is worth separating them. Pooling the conjugate paths—which share the call-ask and put-ask quotes and whose payoffs sum to $-2\text{cost}'$ —makes the distribution symmetric about $-\text{cost}'$ by construction. That it is also sharply *peaked* there, rather than spread across the fair band, is the economic content: a maker who improves one side's MMA lowers the conjugate by the same amount, so the symmetric quote concedes no advantage.

Competitive quoting thus keeps the modal MMA near $-\text{cost}'$, and since the futures spread is small in our data $\text{cost}' \approx \text{cost}$ (Figures 3 and 4). Stale or aggressive quotes populate the tails; the right tail crossing zero is the AMU.

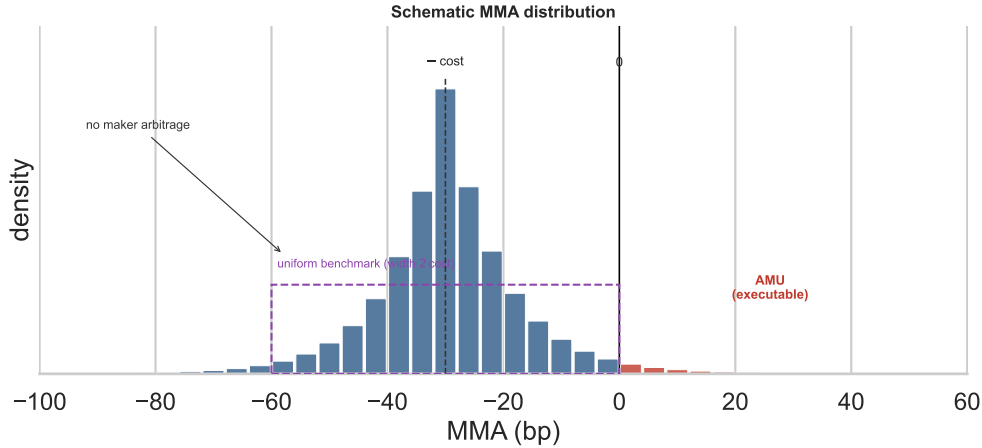


Figure 3: Schematic MMA distribution. The dashed rectangle over $(-2\text{cost}', 0)$ is the no-information benchmark implied by the no-AMU band (uniform quoting), with $\text{cost}' \equiv \text{cost} - \frac{1}{2}(F_{ask} - F_{bid})e^{-rT}$. Competitive quoting instead concentrates the mode at $-\text{cost}'$, the symmetric equilibrium of the conjugate forward/backward paths; the mass that crosses zero into positive territory is the AMU. Schematic, illustrative levels.

3.4 Data and sample

The sample uses high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching-expiry futures from Tardis.dev (Tardis.dev 2026), spanning 2020 through June 2026, with synchronized best bids and asks for calls, puts, and futures. Ticks are processed with our open-source codebases—`balazs14/TardisProject` and `balazs14/publications` (subfolder `AMU-liquidity-premium`)—which implement synchronization, filtering, parity-path construction, and AMU aggregation (Králik 2026 b,a). The same calculation applies across the full horizon, so pre/post-2024 comparisons are coherent.

Table 2 summarizes coverage. Deribit lists a wider surface—more contracts, expirations, and strikes per day—while spreads are broadly comparable, slightly wider for ETH on OKX. The OKX series are longer because Deribit introduced matching-expiry futures later.

3.5 Panel construction

Observations are synchronized on a regular time grid, filtered to matched pairs with usable quotes, and aggregated into cells indexed by (day, exchange, ref_sym, rel_strike_bucket, tte_bucket). This preserves the cross-section while supporting the time-series analysis. Table 3 collects the processing assumptions.

	BTCUSD		ETHUSD	
	deribit	okex	deribit	okex
Avg num contracts	114	50	66	34
Avg num expirations	7	5	7	4
Avg num strikes	62	36	44	32
Avg call spread (bp)	57	56	59	66
Avg put spread (bp)	56	56	59	64
Num days in sample	1139	2269	1139	2145

Table 2: Summary statistics of matched put–call pairs in the 2020–2026 sample.

Table 3: Data-processing assumptions for MMA/AMU construction.

Component	Assumption
AMU clipping rule	MMA series are clipped to $[0, 100]$ basis points before positive-part averaging.
MMA clipping rule	MMA series are clipped to $[-500, +500]$ basis points.
Spread filter	Call and put spreads are capped at 1000 basis points.
Relative-strike filter	Allowed range is $[0.79, 1.21]$.
Minimum quote size	<code>min_quote_size_dollar</code> must be at least 5000.
Subsampling	Last-known observation carried forward to the configured interval (default <code>5min</code>); a leg with a missing or non-updated bid/ask price or size is marked stale.
Stale-quote filter	disabled: when enabled, ticks with a stale call, put, or spot leg are dropped before any AMU/MMA aggregation.

4 The level and cross-section of the absolute market unfairness

4.1 The shape of the MMA distribution and why it peaks below zero

The MMA density peaks at minus the cost.

The most informative object in the data is the shape of the MMA distribution (Figure 4). Pooled across markets and paths, the density peaks sharply at minus the trading cost—near -30 basis points—and falls away on both sides. Zero splits the region where the maker cannot profit from the positive tail that feeds AMU.

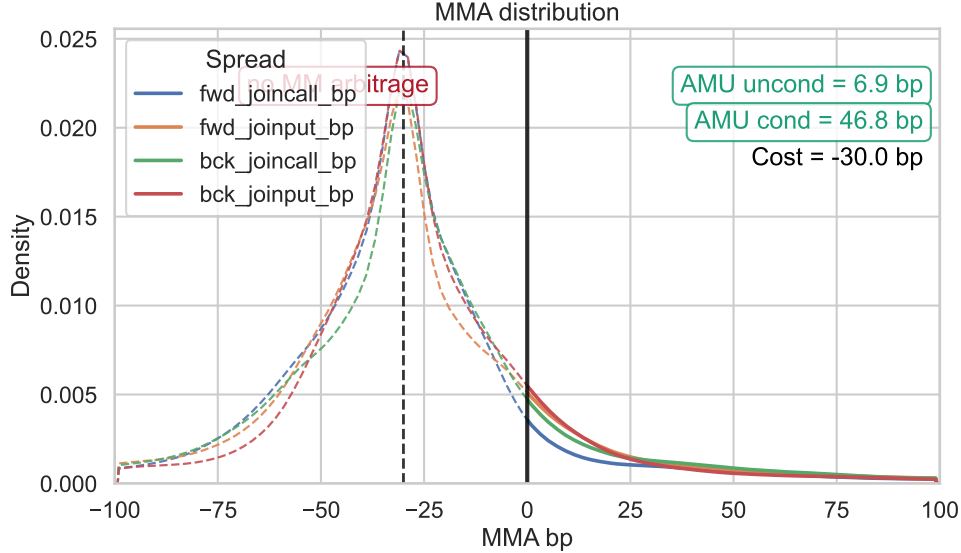


Figure 4: Average density of the four MMA paths across all markets. The distribution peaks at minus the round-trip cost (dashed line); everything to the left of zero yields no maker profit, and only the positive tail to the right of zero contributes to AMU.

The peak is where competition drove the wedge down to cost.

The reading is central to the paper. Makers improve quotes only as much as competition forces, so the mass sits at $-\text{cost}$: in the modal state the wedge is competed down to about the cost of the trade. Quoting is noisy, so the distribution is not a point mass. Quotes too good for the buy side fill the left tail; quotes too bad leave a positive wedge and fill the right tail. The right tail’s conditional mean is the conditional AMU, its share is ϕ , and their product is the unconditional AMU. AMU is nonzero because repricing is not instant: after a shock some quotes are briefly stale or mis-set. Faster, denser competition erases both tails. Everything that follows measures this right tail across venues, the surface, and time.

4.2 RQ1: AMU across exchanges, moneyness, and maturity

Venues differ moderately; the thinnest book has the most AMU.

The global averages (Table 4) show venues differ moderately, not on a simple BTC/ETH split. The standout is ETH on OKX, the thinnest book (Table 2), as expected if AMU tracks quoting quality: the least deep book leaves the largest wedge above cost. Positive MMA is not rare: a large fraction of ticks carry at least one positive-MMA path, so the positive-part average is a routine phenomenon, not isolated glitches. The ratio of the unconditional to the conditional column recovers ϕ (Eq. 5).

		cond	uncond	positive MMA	num pairs
underlying	exchange				
BTCUSD	deribit	47.62	6.71	13,614,478	37,240,737
	okex	46.47	6.64	11,966,194	32,465,782
ETHUSD	deribit	44.30	6.65	8,306,658	21,512,721
	okex	48.24	7.79	8,691,392	21,193,679

Table 4: Global AMU summary across exchanges and underlyings. AMU is the positive-part mean of MMA, in basis points of notional: *cond* averages over the positive-MMA paths only, *uncond* over all paths (equivalently, *cond* times the fraction of paths with positive MMA). A *path* is one of the four put–call-parity join paths (forward/backward $\times$ join-call/join-put) evaluated at a single synchronized quote; each matched put–call pair contributes four paths. The last two columns count ticks: *positive MMA* is the number of synchronized ticks with at least one positive-MMA path, and *num pairs* the number of matched put–call pair ticks.

Over the surface, AMU varies mainly through frequency, not size.

The surface shape is sharper than the venue averages and regular across all four markets. In Figure 5, the positive MMA quote frequency ϕ (lower panel) traces a deep V—near one in the wings, a few percent at the money—while the conditional size (upper panel) is flat. Their product, the unconditional AMU, inherits the V. Figure 6 is the same in maturity with a U-shaped frequency: high at the shortest maturities, minimal near one month, rising again for long-dated contracts. AMU is smallest for month-scale, near-the-money contracts and largest in the illiquid corners. Over the surface it is the *incidence* of mis-quoting that varies, not its size.

Position on the surface matters more than venue.

These regularities answer RQ1. AMU differs across venues, but position on the surface—distance from the money and from a one-month horizon—explains far more of its variation than venue does. Both patterns line up with liquidity: the wedge is largest where books are thin and quotes stale.

5 The 2024 regulatory shift

Regulation, not parity, changed in 2024.

Three milestones cluster in 2024. On January 10 the SEC approved spot Bitcoin ETPs, connecting crypto price risk to regulated custody, authorized-participant arbitrage, and mainstream brokerage (U.S. Securities and Exchange Commission 2024b). In May it approved spot Ether ETFs, which began trading in July (U.S. Securities and Exchange Commission 2024a). In parallel, MiCA came into force in stages: stablecoin provisions in mid-2024, the CASP licensing regime at end-2024 (European Securities and Markets Authority 2024). None of this changed put–call parity; it changed who can hold, finance, and hedge crypto exposure at scale, and under what legal certainty.

Bigger, regulated players plausibly need a smaller premium.

This is where AMU is useful as a diagnostic. Whatever keeps a parity wedge open, it is plausibly cheaper to bear for a large, diversified, well-capitalized institution than for

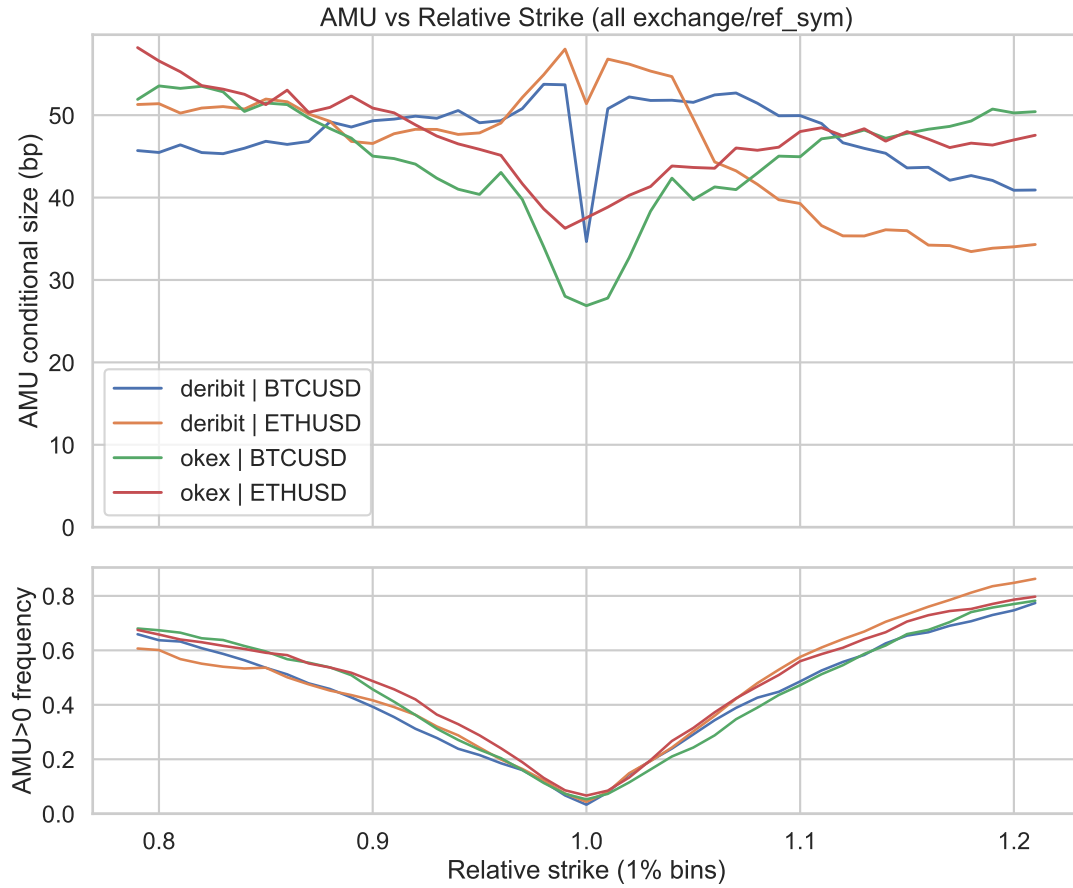


Figure 5: AMU by relative strike across exchanges and underlyings, decomposed into its two margins: the conditional size (upper panel, comparatively flat) and the positive MMA quote frequency ϕ (lower panel, a deep V minimized at the money). Their product, the unconditional AMU, is minimized at the money and rises sharply into the wings.

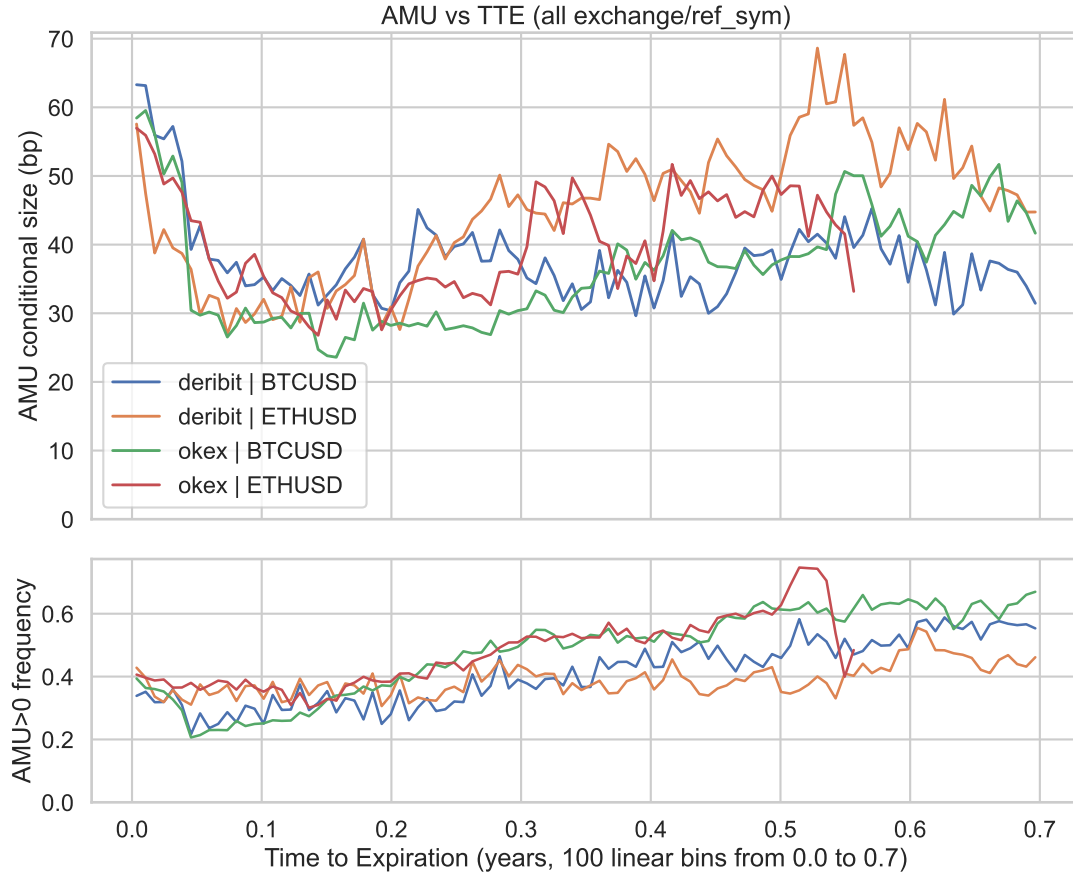


Figure 6: AMU by time to expiry across exchanges and underlyings: the conditional size (upper panel, comparatively flat) and the positive MMA quote frequency ϕ (lower panel). The frequency—and hence the unconditional AMU—is U-shaped: elevated at the shortest and longest maturities, lowest around one month.

the crypto-native firms that dominated before: regulation eases custody and legal risk, balance-sheet depth eases funding risk, diversification eases inventory and convergence risk. If 2024 brought such participants into quoting these markets—directly, or by financing those who do—one would expect the wedge to shrink, beyond any gain in the observable frictions. This is a working hypothesis, not something we identify cleanly: we treat 2024 as a descriptive regime marker and ask whether the evidence fits.

This gives two descriptive questions.

Two descriptive questions follow: did AMU drop around the 2024 changes, and does the drop line up with the observable frictions (spreads, depth, staleness), or persist beyond them? Sections 6 and 7 take these in turn.

6 Did the absolute market unfairness fall in 2024?

6.1 Time-series and distributional evidence

Daily AMU is high before 2024, low after.

The time-series evidence is direct. Figure 7 plots daily AMU by market against the BTCUSD index. Through 2020–2023 AMU is elevated and volatile, spiking in turbulent episodes; from 2024 it drops to a lower, tighter regime. The spikes cluster in stressed markets, when market making is balance-sheet intensive and quote quality poor.

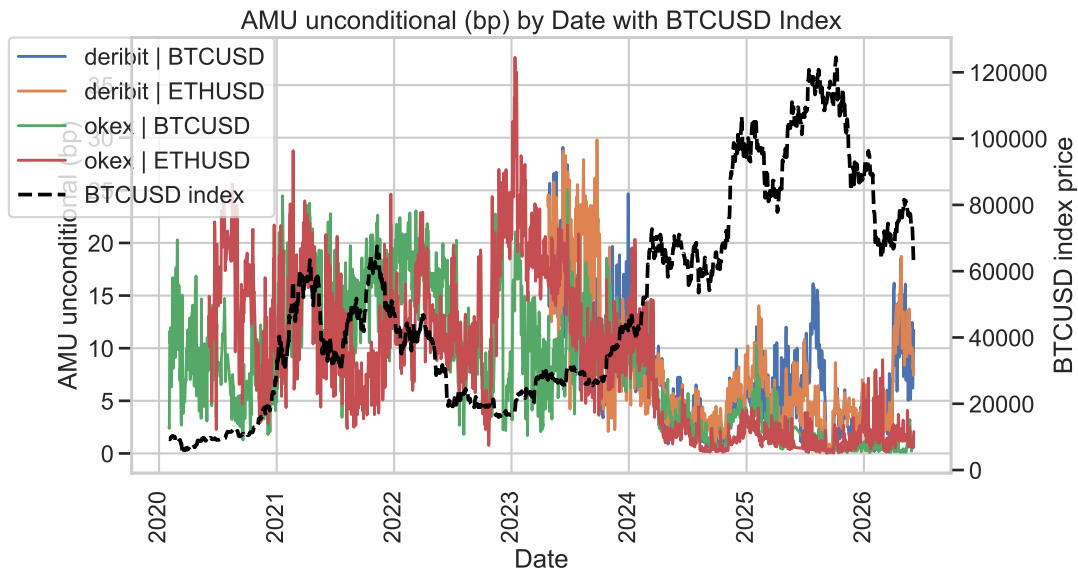


Figure 7: Daily unconditional AMU by exchange and underlying, 2020–2026, with the BTCUSD index overlaid. It is high and volatile before 2024 and compresses to a lower, tighter band afterward.

The step down tracks the 2024 event dates.

Zooming in, Figure 8 tracks the all-market average across 2023–2024 with event dates marked. AMU runs high through 2023, steps down around the January 2024 ETP approval, and keeps falling through the Ether ETF and MiCA milestones. The event-study windows (Figure 9) are noisier—the windows overlap—but show the same tendency: lower and less spiky after events.

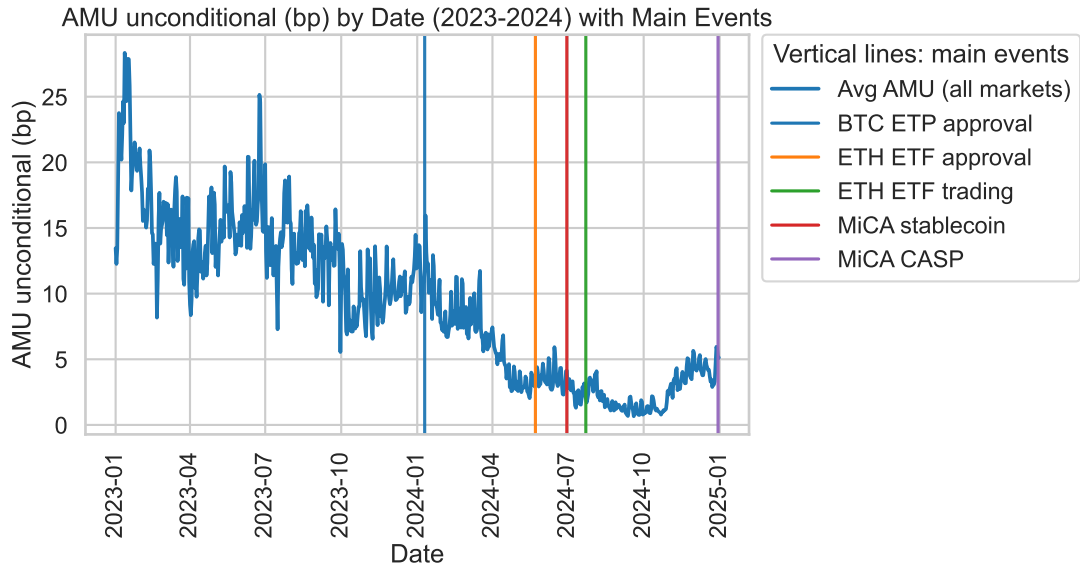


Figure 8: All-market average unconditional AMU, 2023–2024, with the main institutional event dates marked by vertical lines.

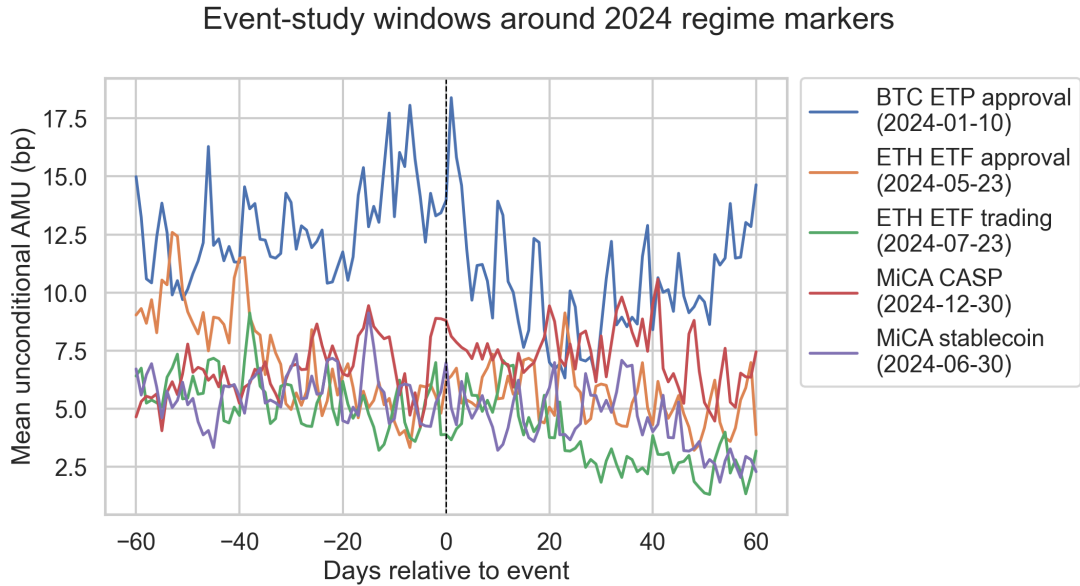


Figure 9: Event-study windows for the unconditional AMU around the main 2024 institutional event dates, shown separately by event.

The whole distribution concentrated at the cost floor.

Two figures show *how* the compression happened. Figure 10 overlays the pre- and post-2024 MMA histograms: the post-2024 density concentrates at the cost floor with thinner tails, and average unconditional AMU falls from 13.1 bp to 5.5 bp. This is consistent with the mechanism of Section 4.1: denser competition erases more of the transient tails. Figure 12 shows the compression over the strike–maturity grid: the pre-2024 hot band at long maturities and away-from-money strikes cools sharply, leaving only the deep-OTM, long-dated corner elevated. Figure 13 shows the change is broad but uneven: conditional AMU fell in every segment, most in the far-expiry bucket, least on Deribit, more for ETH than BTC and for non-ATM than ATM—largest where the pre-2024 wedge was largest.

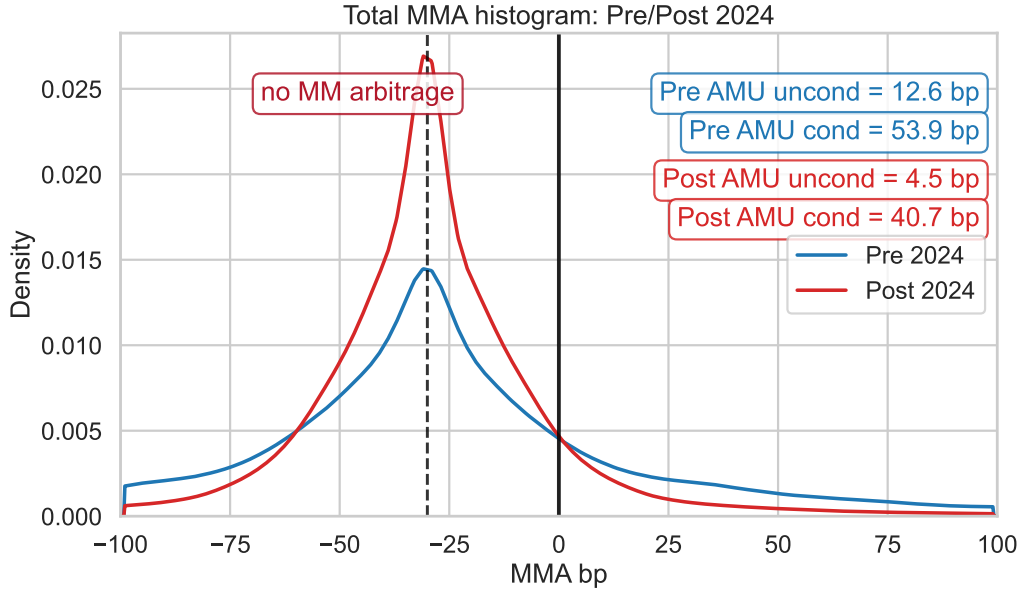


Figure 10: Total MMA histogram across all four paths, split into pre- and post-2024 periods. The post-2024 distribution is more concentrated at the cost floor, and the positive-tail averages—both the conditional and the unconditional AMU, annotated on the figure—fall markedly after 2024.

The drop does not depend on the cost assumption.

The compression is not an artifact of the cost assumption. Figure 11 recomputes both AMU flavours over a grid of costs, pre and post. The pre-to-post drop in unconditional AMU is approximately independent of the assumed cost; the drop in conditional AMU is not.

7 Regression analysis

7.1 Empirical design

The outcome is cell-level unconditional AMU.

The dependent variable in all four regressions is the cell-level unconditional AMU, $AMU_{g,t}^u = \phi_{g,t} AMU_{g,t}^c$ (Eq. 5)—a quote-level, panel-observable object. Regressors enter in three blocks: the post-2024 indicator $Post_t$; segment dummies (exchange, underlying,

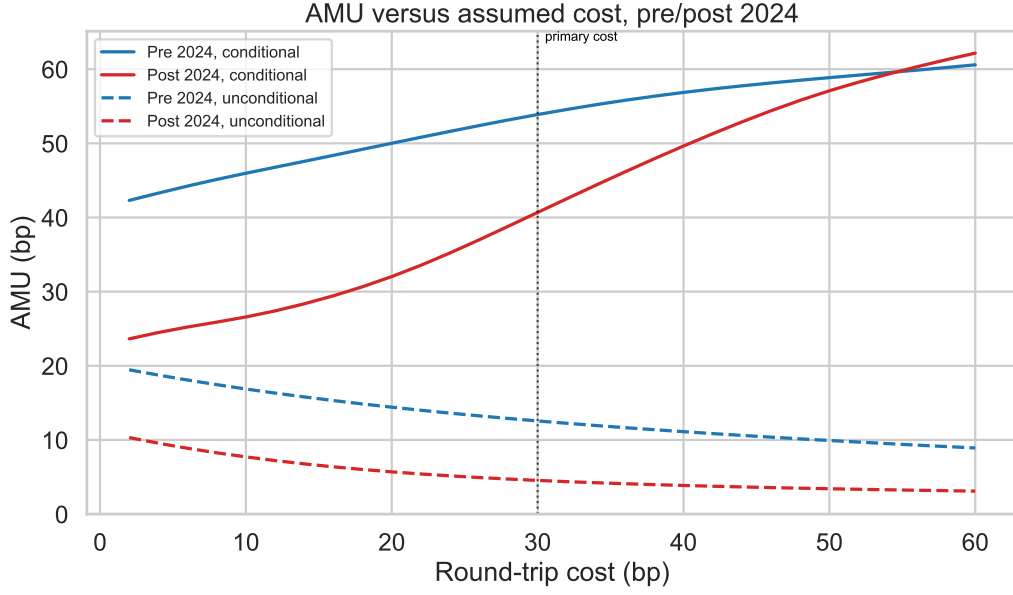


Figure 11: Conditional (solid) and unconditional (dashed) AMU versus the assumed round-trip cost, pre- (blue) versus post-2024 (red), pooled across markets and paths; the dotted line marks the primary cost (30 bp). The pre-to-post gap in unconditional AMU is approximately flat in cost; the gap in conditional AMU shrinks sharply as cost rises. The unconditional measure is therefore the natural object for the regime comparison.

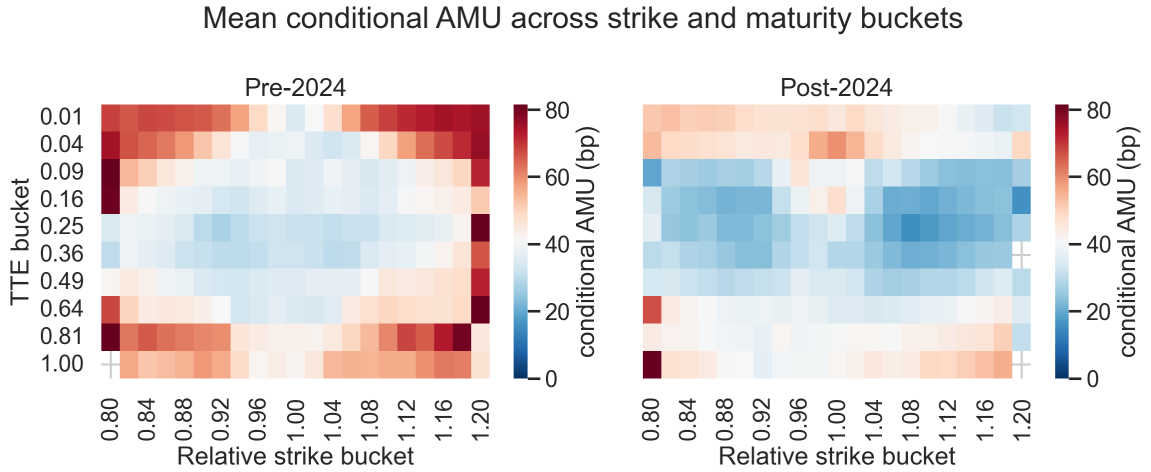


Figure 12: Mean *conditional* AMU across relative-strike and maturity buckets, pre- versus post-2024. The high-AMU interior of the surface cools after 2024; only the deep-OTM, long-dated corner stays elevated.

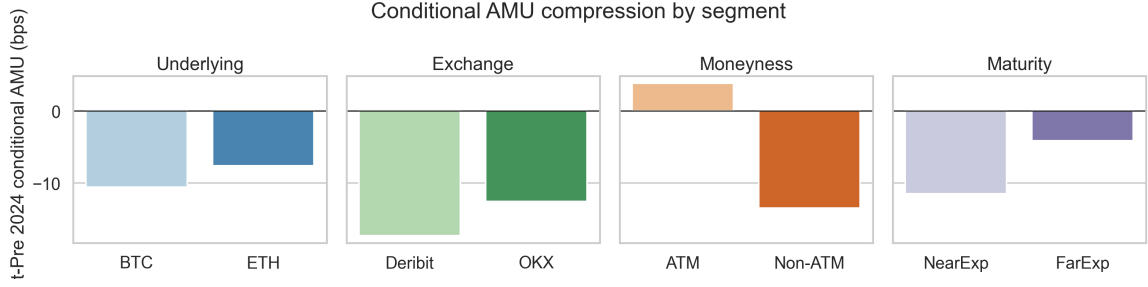


Figure 13: Pre- to post-2024 change in *conditional* AMU by segment (BTC/ETH, Deribit/OKX, ATM/Non-ATM, NearExp/FarExp). Every segment compresses; the far-expiry bucket compresses most.

moneyness, maturity); and three liquidity frictions,

$$\text{Spr}_{g,t} = \frac{1}{2} \left(\text{mean_call_spread_bp}_{g,t} + \text{mean_put_spread_bp}_{g,t} \right), \quad (7)$$

$$\text{Depth}_{g,t} = \log \left(\text{mean_min_quote_size_dollar}_{g,t} \right), \quad (8)$$

$$\text{Stale}_{g,t} = \text{mean stale-quote fraction across call, put, and spot legs.} \quad (9)$$

Frictions enter as descriptors, not to identify a premium.

The direct-cost term is already subtracted inside every path payoff (Table 1), so AMU is the wedge that survives above that floor (Eq. 1). We add the friction variables not to identify a structural premium but to describe: how much of AMU, and of its 2024 change, moves with observable book quality—wider spreads, thinner depth, staler quotes (Amihud & Mendelson 1986, Shleifer & Vishny 1997, Gârleanu & Pedersen 2011, Bollen et al. 2004)—and how much does not.

The break date is January 10, 2024.

Post_t equals one on and after January 10, 2024—the Bitcoin ETP approval, first of the 2024 milestones—treated as a single regime date rather than separate shocks.

We run four nested specifications.

We estimate four specifications, adding regressors in blocks:

$$\text{AMU}_{g,t}^u = \beta_{\text{Post}} \text{Post}_t + \varepsilon_{g,t}, \quad (10)$$

$$\text{AMU}_{g,t}^u = \beta_{\text{Post}} \text{Post}_t + \gamma_g + \varepsilon_{g,t}, \quad (11)$$

$$\text{AMU}_{g,t}^u = \beta_{\text{Post}} \text{Post}_t + \beta_{\text{OKX}} \text{OKX}_g + \beta_{\text{ETH}} \text{ETH}_g + \beta_{\text{ATM}} \text{ATM}_g + \beta_{\text{NE}} \text{NearExp}_g + \varepsilon_{g,t}, \quad (12)$$

$$\text{AMU}_{g,t}^u = \beta_{\text{Post}} \text{Post}_t + \beta_{\text{Depth}} \text{Depth}_{g,t} + \beta_{\text{Spr}} \text{Spr}_{g,t} + \beta_{\text{Stale}} \text{Stale}_{g,t} + \gamma_g + \varepsilon_{g,t}, \quad (13)$$

all with a constant and HC1 standard errors. Specification (1) is the raw regime effect. Specification (2) adds cell fixed effects γ_g , isolating the within-cell shift. Specification (3) instead adds the segment dummies as pooled OLS: there the dummies are the objects of interest (RQ1), so we estimate rather than absorb them. Specification (4) adds the frictions with cell fixed effects. We prefer cell fixed effects because the surface dependence is strongly nonlinear (Figures 5–6); linear dummies would leave segment structure in the residual. In (4), β_{Post} is the within-cell change net of both the frictions and the full segment structure. Table 5 defines the abbreviations.

Table 5: Abbreviations used in the empirical specifications.

Abbreviation	Definition
Post_t	Post-2024 regime indicator (1 for dates on/after 2024-01-10, 0 otherwise).
<i>Liquidity-friction variables</i>	
$\text{Spr}_{g,t}$	Average put–call spread in basis points, $0.5 \times (\text{mean_call_spread_bp} + \text{mean_put_spread_bp})$.
$\text{Depth}_{g,t}$	Log quote-depth proxy, $\log(\text{mean_min_quote_size_dollar})$, min taken over put/call bid/ask quote sizes.
$\text{Stale}_{g,t}$	Mean stale-quote fraction across call, put, and spot legs.
<i>Segment dummies (components of the cell fixed effect γ_g)</i>	
OKX_g	OKX indicator (1 for OKX, 0 for Deribit).
ETH_g	ETH-underlying segment indicator.
ATM_g	At-the-money indicator, $ \text{rel_strike_bucket} - 1 \leq 0.05$.
NearExp_g	Near-expiry indicator, $\text{tte_bucket} \leq 0.35$ years.
γ_g	Cell fixed effect for the tuple (exchange, ref_sym, rel_strike_bucket, tte_bucket).
$\varepsilon_{g,t}$	Idiosyncratic error.

7.2 Results

We use the regressions to check that AMU behaves like a market-quality statistic.

We do not try to identify why AMU fell. We ask whether it behaves as a good market-quality measure should. Two things are required: it should fall when the market is independently believed to have improved (the 2024 institutionalization), and that fall should not be an artifact of which contracts trade. Table 6 checks both. The columns are (1) the raw post-2024 shift; (2) plus cell fixed effects; (3) segment dummies (pooled OLS); (4) the frictions with cell fixed effects. Standard errors are HC1; the *Eff* columns report the standardized effect $\beta \text{std}(X)/\text{std}(Y)$.

AMU is markedly lower after 2024.

Specification (1). The post-2024 coefficient is negative and large. AMU is much lower after 2024—the period the market is widely agreed to have improved. This is the first thing a market-quality measure should show, and answers RQ2.

The drop is within cells, not a change in contract mix.

Specification (2). Adding cell fixed effects barely moves the post-2024 coefficient. The drop happens *within* fixed strike–maturity–venue cells over time, not because the traded mix shifted. This rules out the main composition artifact—the second thing we need.

AMU differs across segments, but only slightly.

Specification (3). With segment dummies (pooled OLS), AMU differs by exchange, underlying, moneyness, and maturity—all significant but small. Venue and underlying move AMU far less than position on the surface. The low R^2 is expected: linear dummies cannot capture the nonlinear surface, which is why (4) uses fixed effects. This answers RQ1.

Most of the drop survives the friction controls.

Specification (4). Adding spreads, depth, and staleness (staleness strongest) shrinks the post-2024 coefficient but does not remove it: a large part of the drop is not accounted for by these observable book-quality measures. The positive tail is, as expected, made of transiently mispriced, slow-to-update quotes. We read the surviving drop as consistent with quoting that improved beyond deeper passive books—but the design is descriptive and identifies neither a mechanism nor a cause. This is our reading of RQ3.

Term	(1) Coef	(2) Coef	(3) Coef	(4) Coef	(1) Eff	(2) Eff	(3) Eff	(4) Eff
Post _t	-8.333*** (0.042)	-9.588*** (0.038)	-9.728*** (0.040)	-1.862*** (0.031)	-30%	-35%	-35%	-7%
OKX _g			-2.836*** (0.035)				-11%	
ETH _g			0.830*** (0.034)				3%	
ATM _g			-8.019*** (0.031)				-28%	
NearExp _g			-0.661*** (0.037)				-2%	
Depth _{g,t}				-0.929*** (0.013)				-14%
Spr _{g,t}				0.097*** (0.000)				66%
Stale _{g,t}				18.837*** (0.235)				11%
Total R^2	0.092	0.321	0.179	0.662				
Observations	495,367	495,367	495,367	495,367				
Fixed effects	None	cell_id	None	cell_id				

Table 6: Regressions of the cell-level unconditional AMU. Column (1) regresses AMU on the post-2024 indicator; (2) adds cell fixed effects to (1), isolating the within-cell regime effect; (3) instead adds the exchange, underlying, moneyness, and maturity dummies (pooled OLS); (4) adds the depth, spread, and staleness controls with cell fixed effects. HC1 standard errors in parentheses; stars denote p -value bands from the t -statistic. The columns headed *Eff* report the standardized effect $\beta \text{std}(X)/\text{std}(Y)$: the change in AMU, in standard deviations of AMU, per one-standard-deviation increase in the regressor.

AMU tracks the spread channel tightly.

Figure 14 shows how closely the wedge co-moves with the spread that specification (4) controls for—which is why the part of the drop that survives it is worth noting.

The regressions corroborate AMU as a market-quality measure.

Taken together: AMU differs significantly but modestly across segments (RQ1); it is much lower after 2024 (RQ2); and most of that drop is within cells and survives the friction

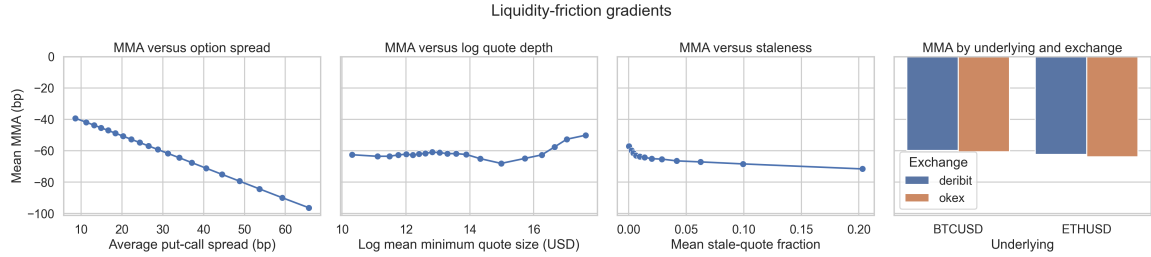


Figure 14: Mean MMA against average put–call spread, log quote depth, and stale-quote fraction, and by underlying and exchange. Mean MMA falls almost one-for-one with the spread and moves with staleness, the friction channels that specification (4) controls for.

controls (RQ3). Because 2024 is independently known as a period of market improvement, AMU moving the way a market-quality statistic should—down, within cells, beyond the obvious frictions—is what gives us confidence in it as a measure. We do not claim to identify which participants drove the change.

8 Conclusion

AMU is a model-free symptom of imperfect quoting.

We propose *Absolute Market Unfairness*: the cost-adjusted put–call-parity violation left in the standing quotes of inverse cryptocurrency option markets. It is model-free and comparable in basis points—an incontrovertible sign that the top of the book is not where frictionless parity would put it—not a full account of a venue’s liquidity or a cleanly identified risk premium. It says nothing about fundamental mispricing and guarantees no dollar profit.

AMU is small across venues, shaped by the surface, and fell in 2024.

Using Deribit and OKX BTCUSD and ETHUSD options from 2020 through June 2026: AMU differs across venues and underlyings, but little next to its variation over the surface, where it is governed mainly by how often a wedge appears—smallest for month-scale, near-the-money contracts, largest in the illiquid corners. It fell sharply around 2024, tracking the ETF and MiCA dates, robustly to the cost level, and most of the drop survives controls for spreads, depth, and staleness within cells. This is consistent with the entry of participants whose quoting improved market quality, not just more passive depth—though the design is descriptive and does not identify the mechanism. The value of AMU is that it turns a qualitative worry about quote quality into a model-free number, comparable across venues, underlyings, and maturities, and trackable through regulatory change.

References

- Alexander, C., Chen, D. & Imeraj, A. (2023), ‘Crypto quanto and inverse options’, *Mathematical Finance* **33**(4), 1005–1043.
- Alexander, C., Deng, J. & Imeraj, A. (2024), ‘Arbitrage opportunities and efficiency tests in crypto derivatives markets’, *Journal of Financial Markets* **71**, 100938.
- Amihud, Y. & Mendelson, H. (1986), ‘Asset pricing and the bid-ask spread’, *Journal of Financial Economics* **17**(2), 223–249.

- Bollen, N. P. B., Smith, T. & Whaley, R. E. (2004), ‘Modeling the bid/ask spread: measuring the inventory-holding premium’, *Journal of Financial Economics* **72**(1), 97–141.
- Chen, S., Chng, M. T. & Liu, Q. (2021), ‘The implied arbitrage mechanism in financial markets’, *Journal of Econometrics* **222**(1), 468–483.
- Cremers, M. & Weinbaum, D. (2010), ‘Deviations from put-call parity and stock return predictability’, *Journal of Financial and Quantitative Analysis* **45**(2), 335–367.
- Du, W., Tepper, A. & Verdelhan, A. (2018), ‘Deviations from covered interest rate parity’, *The Journal of Finance* **73**(3), 915–957.
- El-Mekkaoui, M. & Flood, M. D. (1998), ‘Put-call parity revisited: Intraday evidence from the options market’, *Journal of International Financial Markets, Institutions and Money* **8**(3–4), 357–376.
- European Securities and Markets Authority (2024), ‘Markets in crypto-assets regulation (mica)’, Regulatory information page. MiCA became applicable in stages in 2024. Accessed 2026-06-04.
URL: <https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/markets-crypto-assets-regulation-mica>
- Felföldi-Szűcs, N. (2024), ‘Put-call parity in crypto option markets’, *Finance Research Letters* **61**, 104963.
- Gârleanu, N. & Pedersen, L. H. (2011), ‘Margin-based asset pricing and deviations from the law of one price’, *The Review of Financial Studies* **24**(6), 1980–2022.
- Gromb, D. & Vayanos, D. (2010), ‘Limits of arbitrage’, *Annual Review of Financial Economics* **2**(1), 251–275.
- Kamara, A. & Miller, T. W. (1995), ‘Daily and intradaily tests of european put-call parity’, *The Journal of Financial and Quantitative Analysis* **30**(4), 519–539.
- Klemkosky, R. C. & Resnick, B. G. (1979), ‘Put-call parity and market efficiency’, *The Journal of Finance* **34**(5), 1141–1155.
- Králik, B. (2026a), ‘publications: Amu-liquidity-premium’, GitHub repository. Accessed 2026-07-16.
URL: <https://github.com/balazs14/publications/tree/main/AMU-liquidity-premium>
- Králik, B. (2026b), ‘Tardisproject’, GitHub repository. Accessed 2026-07-16.
URL: <https://github.com/balazs14/TardisProject>
- Lucic, V. & Sepp, A. (2024), ‘Valuation and hedging of cryptocurrency inverse options’, *Quantitative Finance*.
- Nisbet, M. (1992), ‘Put-call parity theory and an empirical test of the efficiency of the london traded options market’, *Journal of Banking & Finance* **16**(2), 381–403.
- Pan, K. & Zeng, Y. (2017), Etf arbitrage under liquidity mismatch, Working Paper 59, European Systemic Risk Board.

- Pontiff, J. (1996), ‘Costly arbitrage: Evidence from closed-end funds’, *The Quarterly Journal of Economics* **111**(4), 1135–1151.
- Pontiff, J. (2006), ‘Costly arbitrage and the myth of idiosyncratic risk’, *Journal of Accounting and Economics* **42**(1–2), 35–52.
- Shleifer, A. & Vishny, R. W. (1997), ‘The limits of arbitrage’, *The Journal of Finance* **52**(1), 35–55.
- Tardis.dev (2026), ‘Tardis: Historical crypto market data api’, Product and documentation website. Accessed 2026-06-16.
URL: <https://tardis.dev/>
- Todorov, K. (2021), ‘The anatomy of bond etf arbitrage’, *BIS Quarterly Review* pp. 41–53.
- U.S. Securities and Exchange Commission (2024a), ‘Order granting accelerated approval of proposed rule changes to list and trade shares of ether-based exchange-traded products’, SEC Release, May 2024. Accessed 2026-06-04.
URL: <https://www.sec.gov/>
- U.S. Securities and Exchange Commission (2024b), ‘Statement on the approval of spot bitcoin exchange-traded products’, SEC Chair Statement, January 10, 2024. Accessed 2026-06-04.
URL: <https://www.sec.gov/newsroom/speeches-statements/gensler-statement-spot-bitcoin-011023>